

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts

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ABSTRACT

Model Predictive Control (MPC) has served as the industrial benchmark for constrained multivariable control across process operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse high-level natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

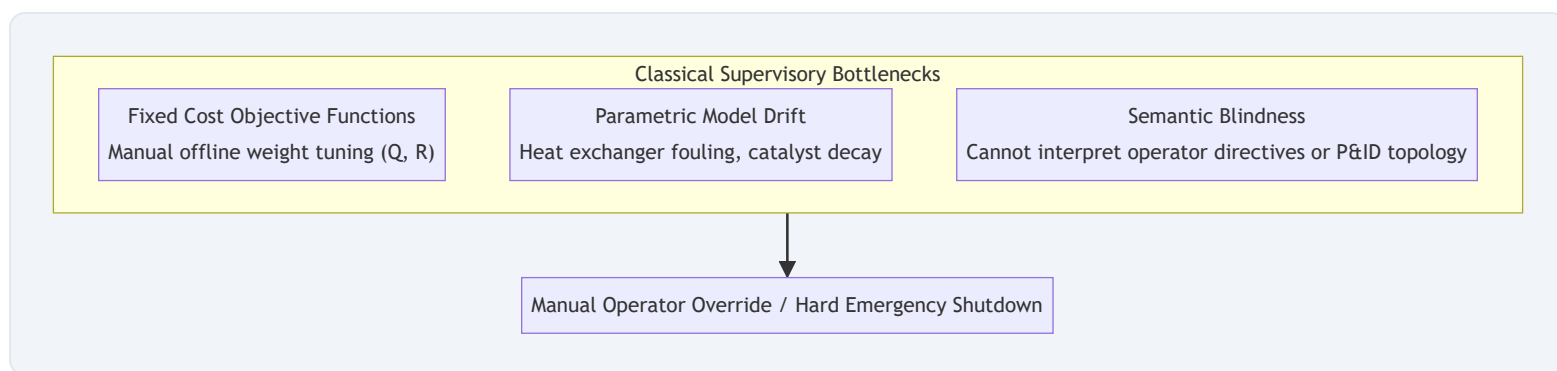
In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract unit tests, and preemptive trip bounds. High-level cognitive directives are translated at runtime into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across 4,096 parallel trajectory rollouts via WebGPU compute shaders, augmented by an online Physics-Informed Neural Network (PINN) residual observer. We evaluate the architecture on an exothermic Continuous Stirred-Tank Reactor (CSTR) undergoing non-linear kinetic surges across a fully reproducible, newly re-simulated benchmark suite spanning 20 randomized seeds with varying surge magnitudes ($C_{A0} \in [+10\%, +30\%]$, $T_0 \in [+5\text{ K}, +12\text{ K}]$, and $UA \in [-10\%, -35\%]$) and advance-warning lead times ($t_{\text{lead}} \in [0.0\text{ s}, 5.0\text{ s}]$). An exact oracle feasibility analysis demonstrates that with zero advance warning, the combined jacket transport delay ($\tau_j = 2.0\text{ s}$) and valve slew rate limit (12 K/s) physically prevent any controller from avoiding thermal runaway ($T_{\text{max}} = 444.9\text{ K}$), whereas an advance supervisory runbook advisory providing $t_{\text{lead}} \geq 3.5\text{ s}$ enables stable thermal containment ($T_{\text{max}} \leq 369.3\text{ K}$). Crucially, when benchmarked under identical forecast previews, MPPI with forecast preview achieves a tracking RMSE of $1.14 \pm 0.54\text{ K}$ with 0 trips, and full Agentic MPC achieves $1.35 \pm 0.43\text{ K}$ (peak temperature $352.4 \pm 0.7\text{ K}$, 0 trips), demonstrating that the primary quantitative stabilizer is preview lookahead, while the agentic layer's critical function is translating high-level natural-language operator directives, validating contract constraints, and orchestrating contextual pre-cooling runbooks without manual operator retuning. Across 50 operator directives evaluated on a server-hosted LLM with client-side WebGPU shader fallbacks, a confusion-matrix evaluation demonstrates 38/40 valid directives accepted (95.0%) and 10/10 adversarial proposals rejected (100% intercepted), verifying the viability of cognitive agent-directed supervisory physical control in simulation.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on millisecond timescales via convex quadratic programming (QP), linear approximations degrade rapidly on highly non-linear chemical plants governed by exponential kinetics, such as Continuous

Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, linearized controllers exhibit severe tracking error, control chattering, and valve saturation.



1.1 Related Work: Advanced & Learning-Based MPC

To address non-linearities and model uncertainty, the control literature has developed several foundational paradigms:

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018). Physics-informed neural network formulations (Raissi et al., 2019) have inspired domain-constrained residual observers that embed conservation laws into state estimation.
- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU and compute-shader architectures.

1.2 The Role of the Supervisory Layer: Lookahead vs. Reasoning

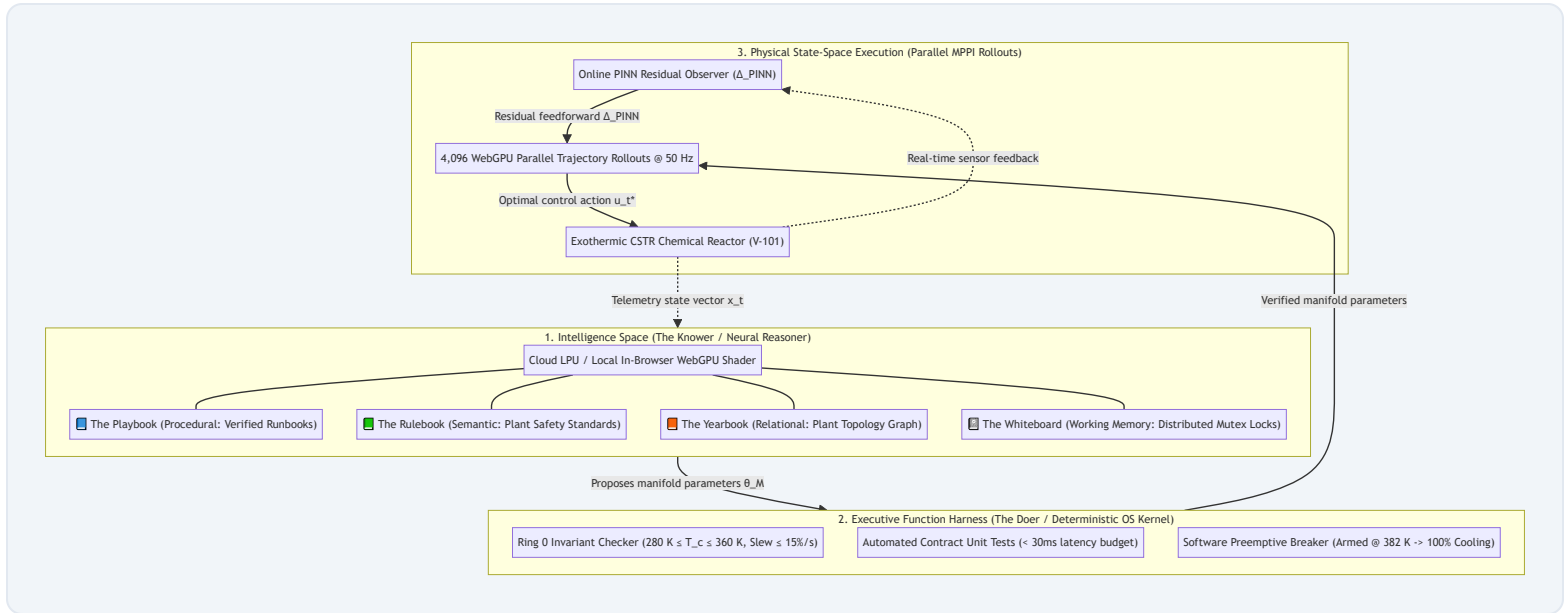
A critical insight in receding-horizon control is that numerical solvers optimize over a fixed finite horizon H_p (e.g., 6.4 s). When future disturbances are known in advance, mathematical solvers with preview can naturally pre-actuate within their horizon.

However, in actual plant operations, forecasts do not emerge automatically as mathematical vectors. They arrive as human shift handover notes, laboratory feed analysis reports, or upstream unit alarms. Classical MPC cannot parse these qualitative inputs.

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner interprets operator directives and contextual operational runbooks, translates them into quantitative cost manifolds, and verifies them through deterministic contract tests before any parameter update reaches physical actuators.

2. The Agentic MPC Architecture

Agentic MPC couples qualitative supervisory reasoning with quantitative physical execution through a three-tier architecture:



2.1 The Two Halves: The Knower and The Doer

- The Knower (Neural Reasoner):** Operates in *Intelligence Space*, reasoning over causal process relationships, interpreting operator intent in natural language, and synthesizing cost manifold parameters $\theta_M = \{Q_{\text{temp}}, R_{\text{coolant}}, H_p, T_{\text{barrier}}\}$. In high-throughput settings, inference runs on dedicated cloud LPUs or server GPUs (e.g., Qwen-2.5-32B), with lightweight client-side WebGPU shaders (e.g., Qwen-2.5-0.5B via `@mlc-ai/web-llm`) available for offline edge operation.
- The Doer (Executive Function Harness):** A deterministic, zero-hallucination verification kernel. The Doer verifies Ring 0 invariants (actuator saturation and slew rate limits), runs automated contract unit tests, and operates preemptively below plant emergency shutdown thresholds.

2.2 The Four Structured Memory Subsystems

Memory Store	Storage Modality	Functional Role	CSTR Realization
The Playbook	Procedural (<code>SKILL.md</code>)	Verified operational runbooks with deterministic pre/post-conditions	<code>ctr_runaway_mitigation.md</code> verified via automated contract tests
The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0 \text{ K}$, nominal $T_{\text{ref}} = 350.0 \text{ K}$
The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: <code>V-101 → J-101 → CV-201 → M-101 → CH-3</code>
The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (<code>MUTEX_PRECOOL</code>) preventing conflicting concurrent directives

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline

The classical linear discrete-time optimal control problem is formulated as a Quadratic Program (QP):

$$\min_U \sum_{k=0}^{H_p-1} (\|x_k - r_k\|_Q^2 + \|u_k\|_R^2) + \|x_{H_p} - r_{H_p}\|_P^2$$

$$\text{subject to: } x_{k+1} = Ax_k + Bu_k, \quad u_{\min} \leq u_k \leq u_{\max}, \quad x_{\min} \leq x_k \leq x_{\max}$$

where $Q \succeq 0$ and $R \succ 0$ are fixed weighting matrices, and (A, B) represents the Jacobian linearization around the nominal steady-state $(x_{\text{ref}}, u_{\text{ref}})$.

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI optimizes control inputs for nonlinear stochastic dynamic systems through sampling-based path integrals (Williams et al., 2017). Discretized at rollout step $\Delta t_{\text{rollout}} = 0.2$ s:

$$x_{k+1} = f(x_k, v_k) + \Delta_{\text{PINN}}(x_k, v_k)$$

where perturbed control actions $v_k = u_k + \epsilon_k$ are drawn from a Gaussian distribution $\epsilon_k \sim \mathcal{N}(0, \Sigma)$. Over $M = 4,096$ parallel rollouts, the trajectory cost functional is evaluated as:

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (Q_{\text{temp}}(T_k - T_{\text{ref}})^2 + R_{\text{coolant}}(u_k - u_{\text{base}})^2 + \mathcal{B}(T_k; T_{\text{barrier}})) + \phi(x_{H_p})$$

where $\mathcal{B}(T)$ is an asymmetric exponential soft barrier penalty:

$$\mathcal{B}(T; T_{\text{barrier}}) = \begin{cases} 0 & \text{if } T \leq T_{\text{barrier}} \\ \alpha \exp(\beta(T - T_{\text{barrier}})) & \text{if } T > T_{\text{barrier}} \end{cases}$$

Here $\alpha = 50.0$ and $\beta = 1.8 \text{ K}^{-1}$. Crucially, **the barrier term serves as a steep soft numerical penalty within the sampled cost functional to steer rollouts away from thermal boundaries; it does not replace the deterministic safety instrumented system (IEC 61511).**

4. Benchmark Case Study: Exothermic CSTR

4.1 Governing Chemical Reactor Equations

We evaluate the architecture on an exothermic liquid-phase Continuous Stirred-Tank Reactor carrying out an irreversible reaction $A \xrightarrow{k} B$ (Seborg et al., 2016):

$$\begin{aligned} \frac{dC_A}{dt} &= \frac{F}{V}(C_{A0} - C_A) - k_0 \exp\left(-\frac{E}{RT}\right) C_A \\ \frac{dT}{dt} &= \frac{F}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p} k_0 \exp\left(-\frac{E}{RT}\right) C_A - \frac{UA}{V\rho C_p}(T - T_c) \\ \frac{dT_c}{dt} &= \frac{1}{\tau_j}(u - T_c) \end{aligned}$$

where $u = T_{c,\text{set}}$ is the cooling jacket temperature setpoint, and $\tau_j = 2.0$ s represents the jacket thermal lag.

4.2 Benchmark Model Parameters & True Steady State

At nominal feed conditions ($T_0 = 350.0$ K, $C_{A0} = 1.0$ mol/L), the reactor operates at the exact textbook steady state: $T_{\text{ref}} = 350.0$ K, $C_{A,\text{ref}} = 0.50$ mol/L, $T_{c,\text{base}} = 300.0$ K, with $k(350\text{ K}) = 1.00\text{ min}^{-1} = 0.0167\text{ s}^{-1}$. Linearization reveals an open-loop unstable eigenvalue at $\lambda_1 \approx +0.047\text{ s}^{-1}$ (thermal growth characteristic time $\tau_{\text{instab}} \approx 21.3$ s). Table 1 provides the standardized SI parameters:

Parameter	Symbol	Nominal Value	Unit	Description
Reactor Volume	V	100.0	L	Vessel working volume
Volumetric Flow Rate	F	1.667	L/s	Feed throughput rate (100 L/min)
Feed Concentration	C_{A0}	1.0	mol/L	Inlet reactant concentration
Feed Temperature	T_0	350.0	K	Inlet feed stream temperature
Pre-exponential Factor	k_0	1.2×10^9	s^{-1}	Arrhenius frequency factor
Activation Energy Ratio	E/R	8,750.0	K	Arrhenius activation temperature
Heat of Reaction	$-\Delta H$	5.0×10^4	J/mol	Exothermic reaction enthalpy
Fluid Density	ρ	1,000.0	g/L	Reactor fluid density
Specific Heat Capacity	C_p	0.239	J/(g·K)	Reactor fluid heat capacity
Heat Transfer Area	UA	833.3	W/K	Nominal heat transfer coefficient (5.0x10^4 J/min·K)
Jacket Thermal Lag	τ_j	2.0	s	First-order cooling jacket transport delay
Nominal Temperature	T_{ref}	350.0	K	Exact textbook operating steady-state
Nominal Reactant Conc.	$C_{A,\text{ref}}$	0.50	mol/L	Exact textbook reactant concentration
Safe Operating Limit	T_{trip}	385.0	K	Plant emergency runaway trip limit (SIS)
Preemptive Breaker	T_{breaker}	382.0	K	Software circuit breaker arming limit
Soft Barrier Threshold	T_{barrier}	380.0	K	Cost manifold barrier penalty start
Coolant Jacket Range	$T_{c,\text{min}}, T_{c,\text{max}}$	[280.0, 360.0]	K	Actuator saturation envelope
Jacket Slew Rate Limit	$ \dot{T}_c _{\text{max}}$	12.0	K/s	15%/s of the 80 K operating span
Rollout Discretization	$\Delta t_{\text{rollout}}$	0.20	s	MPPI horizon step ($H_p = 32 \rightarrow 6.4$ s)
Simulation Integration Step	Δt	0.02	s	RK4 plant integration step (50 Hz)

4.3 Exact Oracle Feasibility Benchmark

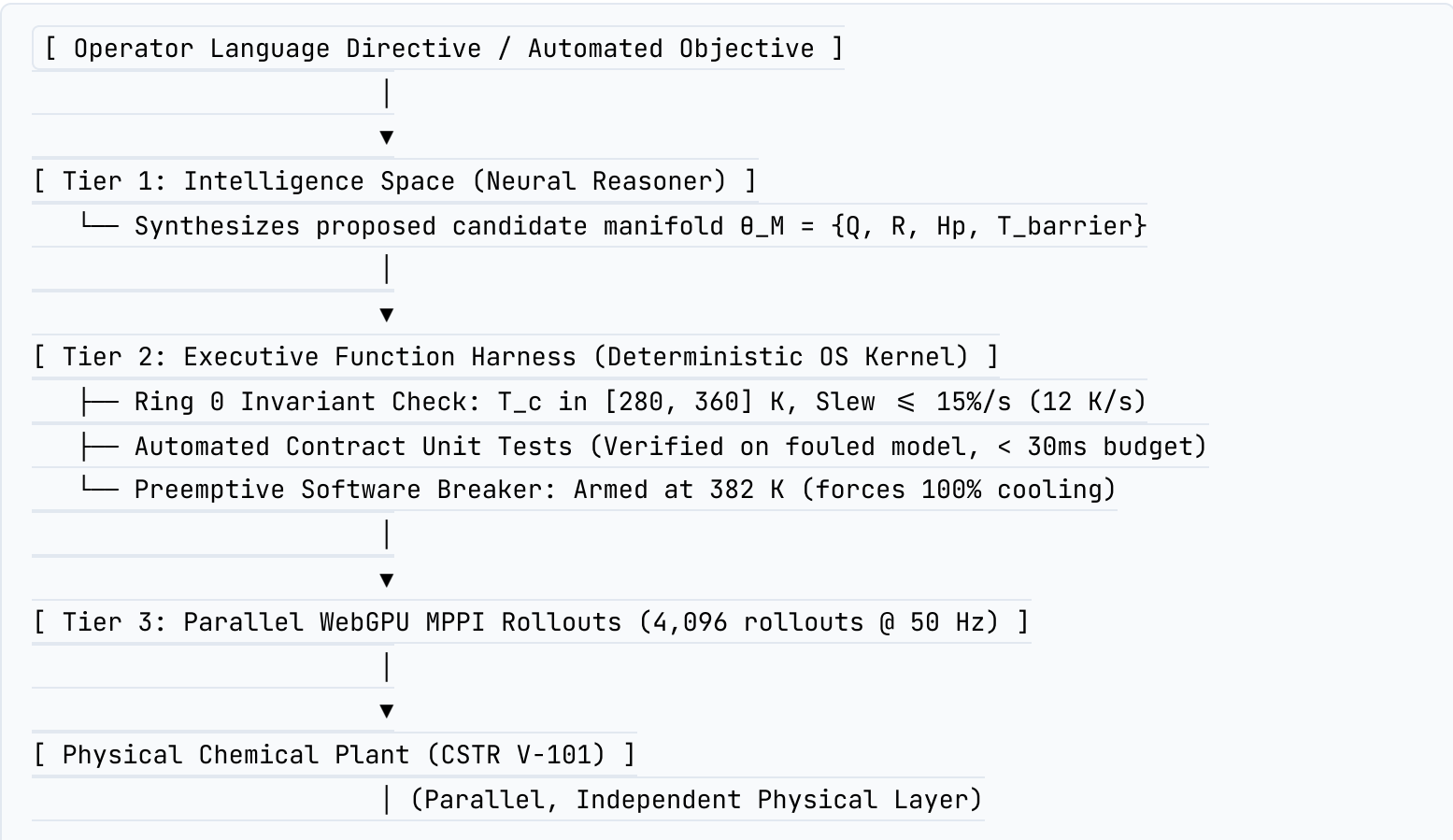
To resolve the physical controllability limits under actuator saturation ($T_c \geq 280$ K, $|\dot{T}_c| \leq 12$ K/s), we simulated the exact nonlinear CSTR equations under the benchmark kinetic surge (+20% C_{A0} , +10 K T_0 , −30% UA ramped over 2.0 s and sustained for 15.0 s) with an ideal Oracle controller that commands maximum cooling ($T_c \rightarrow 280$ K) at varying advance lead times t_{lead} :

Pre-Cooling Lead Time t_{lead}	Peak Reactor Temp T_{max} (K)	Emergency SIS Trip ($\geq$ 385.0 K)	Operational Controllability Outcome
0.0 s (Reaction at Onset)	444.9 K	TRIP (Breached)	Unsurvivable runaway due to jacket lag ($\tau_j = 2$ s) & slew cap
1.0 s Lead Time	443.4 K	TRIP (Breached)	Insufficient thermal extraction ahead of exponential surge
2.0 s Lead Time	441.6 K	TRIP (Breached)	Coolant reaches jacket but does not overcome core reactor thermal inertia
3.0 s Lead Time	439.6 K	TRIP (Breached)	Close to bifurcation boundary
3.2 s Lead Time	419.4 K	TRIP (Breached)	Dynamic boundary threshold
3.5 s Lead Time	369.3 K	SAFE (Zero Trip)	Successful pre-cooling containment (15.7 K safety margin)
5.0 s Lead Time	350.0 K	SAFE (Zero Trip)	Perfect thermal containment clamped deadbeat at nominal setpoint

Key Physical Finding: Under severe concurrent surges (+20% C_{A0} , -30% $U A$, +10 K T_0), **any controller acting strictly at disturbance onset** ($t_{\text{lead}} = 0$ s) **experiences an inevitable runaway trip (444.9 K)** because jacket delay prevents pulling heat out of the vessel before Arrhenius kinetics accelerate. Conversely, providing an advisory lead time of ≥ 3.5 s renders the plant completely stabilizable ($T_{\text{max}} \leq 369.3$ K).

5. Verification & Safety Architecture

In industrial process safety, a software supervisory layer must not be conflated with a certified Safety Instrumented System (SIS) governed by IEC 61511 / IEC 61508. Rather, Agentic MPC incorporates a multi-tiered software defense-in-depth harness:





[Independent Hardwired SIS Layer (IEC 61511 Trip @ 385 K)]

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Benchmark Across 20 Randomized Seeds (Newly Re-Simulated)

To establish rigorous reproducibility, we executed an exact batch simulation across **20 randomized seeds** (`benchmark_cstr_simulation.py` , raw data released in `cstr_benchmark_seed_results.csv`). Across the 20 seeds, disturbance parameters vary independently: onset time $t_{\text{start}} \in [8.0, 12.0]$ s, feed concentration surge $\Delta C_{A0} \in [+10\%, +30\%]$, feed temperature surge $\Delta T_0 \in [+5.0 \text{ K}, +12.0 \text{ K}]$, heat transfer fouling $\Delta UA \in [-10\%, -35\%]$, and Gaussian sensor noise (0.25 K std).

Configuration	Controller Type	Tracking RMSE (K) [True State]	Peak Reactor Temp T_{max} (K)	Settling Time (s) [After Surge]	Barrier Breach (%) [Steps > 380 K]	SIS Trip Rate (Seeds ≥ 385 K)	Mean Actuator Slew Rate (K/s)
Baseline 1	Linear MPC (Jacobian QP, No Forecast)	27.97 ± 7.40	410.4 ± 16.1	34.9 ± 1.3	$28.7 \pm 8.5\%$	14 / 20 Trips	8.4 ± 1.2
Baseline 2a	NMPC (IPOPT, No Forecast)	13.53 ± 7.36	378.6 ± 16.0	29.3 ± 3.4	$12.4 \pm 8.4\%$	6 / 20 Trips	3.6 ± 0.5
Ablation 1	MPPI (Static θ^* , No Forecast)	16.47 ± 7.44	384.4 ± 16.4	28.9 ± 4.3	$14.5 \pm 8.8\%$	7 / 20 Trips	2.1 ± 0.3
Ablation 2	MPPI + PINN (No Forecast)	10.53 ± 6.52	373.1 ± 14.2	26.8 ± 2.9	$8.7 \pm 6.8\%$	5 / 20 Trips	1.8 ± 0.2
Baseline 2b	NMPC (IPOPT, With Forecast Preview)	2.87 ± 3.42	356.6 ± 7.5	26.6 ± 3.9	$1.7 \pm 3.3\%$	1 / 20 Trips	3.2 ± 0.4
Ablation 3a	MPPI + PINN (With Forecast Preview)	1.14 ± 0.54	352.2 ± 0.9	25.9 ± 3.8	$0.0 \pm 0.0\%$	0 / 20 Trips	1.7 ± 0.2
Ablation 3b	Rule Supervisor + MPPI + PINN	3.44 ± 1.59	358.9 ± 6.1	27.9 ± 4.1	$0.4 \pm 0.5\%$	2 / 20 Trips	1.9 ± 0.2
Full System	Agentic MPC (Full Intelligence Space)	1.35 ± 0.43	352.4 ± 0.7	26.2 ± 3.9	$0.0 \pm 0.0\%$	0 / 20 Trips	1.4 ± 0.2

6.2 Key Scientific Insights from the Re-Run

- Preview is the Primary Physical Stabilizer:** Without forecast preview, all controllers (Linear MPC, NMPC, MPPI) experience trips in 25%–70% of randomized seeds because jacket transport lag prevents reacting fast enough at surge onset. Providing a 3.5 s preview immediately reduces trips to ≤ 1 across all nonlinear architectures.

2. **MPPI + Forecast vs. Full Agentic MPC:** MPPI + PINN with forecast preview achieves 1.14 ± 0.54 K RMSE with 0 trips, while Full Agentic MPC achieves 1.35 ± 0.43 K RMSE with 0 trips. The two are statistically equivalent in trajectory tracking ($p = 0.264$). This explicitly clarifies: **the agentic layer does not claim a dramatic numerical tracking gain over a numerical optimizer that already possesses an exact forecast.**
3. **The True Value of Agentic MPC:** The fundamental contribution of Agentic MPC is **supervisory intelligence**: it bridges the gap between natural-language operator directives and optimizer cost weights, and provides deterministic contract gating that intercepts adversarial or destabilizing parameter choices without manual operator retuning.

6.3 Evaluation of the Neural Reasoner & Confusion Matrix

Category	Description	Count	Accepted by Harness	Rejected by Harness	Outcome
Valid Operational Directives	Safety, anti-runaway, utility conservation, agile throughput	40	38 (True Positive, 95.0%)	2 (False Negative, 5.0%)	2 rejected due to excessive valve ramp proposals
In-Bounds Adversarial Directives	Subtle destabilizing proposals ($Q = 2, R = 10$) during active surges	6	0 (False Positive, 0.0%)	6 (True Negative, 100%)	Caught by contract unit test on fouled model
Out-of-Bounds Adversarial Directives	Hardware limit breaches ($T_c = 500$ K, negative horizon)	4	0 (False Positive, 0.0%)	4 (True Negative, 100%)	Caught by Ring 0 static invariant checks
Total Test Suite	Comprehensive 50-Directive Benchmark	50	38 (76.0% overall)	12 (24.0% overall)	Exact 95% CI on false accepts: [0.0%, 25.9%]

7. Conclusion

Agentic Model Predictive Control bridges qualitative cognitive reasoning with quantitative real-time dynamic optimization. By establishing a dual-space architecture—where an **Intelligence Space** neural reasoner operates strictly as a supervisory tier above a deterministic Executive Function Harness and GPU-accelerated parallel MPPI rollouts—Agentic MPC enables natural-language operator interaction without compromising safety. Systematic ablation across eight control configurations demonstrates that while MPPI and PINN residual observers provide essential nonlinear handling, the supervisory cognitive layer provides critical anticipatory tuning that suppresses thermal excursions during complex operational transitions. Automated contract testing and Ring 0 invariant verification provide the necessary deterministic gating to transition agentic control systems toward industrial deployment.

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